Leveraging Context for Multimodal Fallacy Classification in Political Debates

Alessio Pittiglio

DISI, University of Bologna, Italy

Brief introduction

- **Task**: classify a given statement into one of six fallacy types
- Many approaches focus on the statement itself
- Central question…

How can we leverage the ***context*** of a debate to better classify logical fallacies?

Moving beyond the sentence

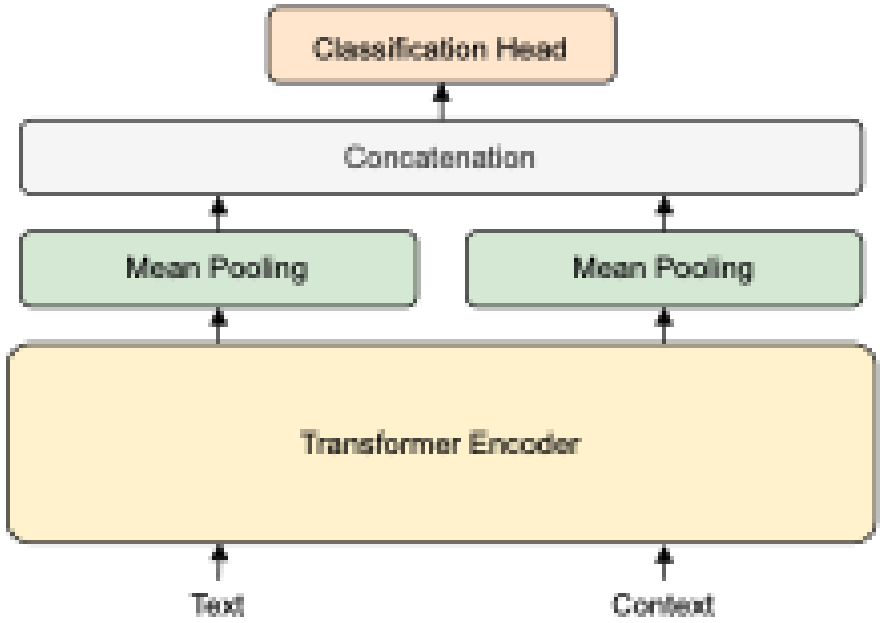
- **Goal**: Integrate context from **text** and **audio**
- **Context:** Preceding sentences + corresponding audio
- **My hypothesis**: More context, *used in the right way* → better performance

Text modality

Backbones: RoBERTa-large, DeBERTaV3-large, ModernBERT-large

1. **Concat**: `[text][SEP][context]`
2. **ContextPool**: Shared encoder + mean pooling
3. **CrossAttn**: Text queries context via gated fusion and attentive pooling

Best: ContextPool + RoBERTa-large



ContextPool

Impact of textual context

Architecture	N=0 (No Ctx)	N=1	N=2	N=3	N=4	N=5	N=6
Transformer (No Context)	0.6131	-	-	-	-	-	-
Concat	-	0.5538	0.5932	0.4767	0.5331	0.5941	0.5431
ContextPool	-	**0.6636**	0.6479	0.5786	**0.6983**	0.6542	**0.6494**
CrossAttn	-	0.5699	0.6086	0.5395	0.6304	0.6344	0.6219
w/ Gate	-	0.5867	0.6171	0.5487	0.5388	0.6181	0.6032
w/ Attentive Pooling	-	0.6603	**0.6527**	0.6227	0.6590	**0.6579**	0.6383
w/ Gate & Attentive Pool	-	0.6280	0.6520	**0.6407**	0.6261	0.3701	0.6377

Finding 1: Text context is crucial, but there is a *sweet spot!*

Audio modality

- Classify fallacies just from the voice
- **Backbone**: fine-tuned HuBERT Base
- Integrating context isn’t as straightforward as with text
- Adapted the ContextPool idea into an architecture I call **TemporalAvg**

Impact of audio context

Architecture	N=0 (No Ctx)	N=1	N=2	N=3	N=4	N=5	N=6
Hubert Base fine-tuned	0.48061	-	-	-	-	-	-
TemporalAvg	-	0.4282	0.4149	0.3856	0.4460	0.4553	0.4518

Finding 2: adding audio context did **not** consistently improve performance

Fusion strategy

Goal: Combine best text and audio models

Method 1: Weighted Average

- ContextPool-RoBERTa + HuBERT-Base fine-tuned
- Weights optimized via *Bayesian optimization*

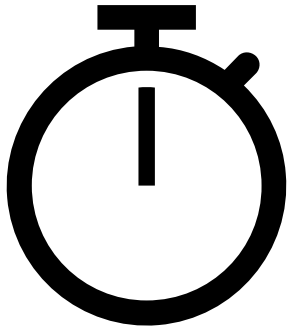
Method 2: Majority Voting (final submission)

Results

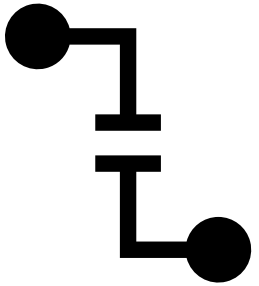
Input	Team	F1
Text-Only	Team NUST	0.4856
	Baseline BiLSTM	0.4721
	My result	**0.4444**
Audio-Only	**My result**	**0.3559**
	Team EvaAdriana	0.1858
	Team Nust	0.1588
Text-Audio	Team Nust	0.4611
	My result	**0.4403**
	Baseline RoBERTa + WavLM	0.3816

Finding 3: Late fusion fails to improve on the text-only model

Limitations and Future Work



Audio truncation.
Affects 17% of data, limiting context for long fallacies (e.g., Slippery Slope).



Late fusion.
Doesn't capture cross-modal interaction (what is said vs. how it's said)

Takeaways

1. Text context is **critical**. ContextPool is a simple yet effective method.
2. Fine-tuned **HuBERT** captures **informative features** for fallacy classification. But context was not helpful.
3. Effective **multimodal fusion** is hard. It's the key challenge for future work!

Check out my code on GitHub:



github.com/alessiopittiglio/mm-argfallacy

Thank you!

Alessio Pittiglio
alessio.pittiglio@studio.unibo.it